

***One small cloud
misconfiguration can expose an
entire production environment***

What is IaC?

(Infrastructure as Code)

- Managing infrastructure using code instead of manual configuration
- Infrastructure becomes:
- Version-controlled
- Automated
- Repeatable
- Commonly used for:
- Cloud resources
- Networks
- Kubernetes clusters
- Storage & permissions

Why IaC is Important in Modern DevOps

- Faster infrastructure deployment
- Automation at scale
- Consistency across environments
- Easy collaboration and recovery

The Real Problem

(Why IaC Security Matters)

- Infrastructure is now written as code
- If insecure configurations are deployed:
- Entire cloud environments become vulnerable
- Common risks:
 - Public S3 buckets
 - Open security groups
 - Weak IAM permissions
 - Exposed databases

What is IaC Security?

- Scanning infrastructure code for security misconfigurations before deployment
- Goal:
- Prevent insecure cloud resources from going live
- “Shift Left Security for Infrastructure”

What IaC Security Tools Detect?

- Misconfigured cloud resources
- Overly permissive access
- Public exposure risks
- Compliance violations
- Weak encryption settings

Policy-as-Code

- Security rules written as code
- Infrastructure automatically validated against policies
- Example:
 - “No public database allowed”
 - “Encryption must be enabled”

Shift Left Infrastructure Security

Real-World Importance

Industry-Standard IaC Security Tools

- Checkov
- Terrascan
- Used to scan:
- Terraform
- Kubernetes configs
- Cloud infrastructure code

Typical IaC Security Workflow

- Developer writes infrastructure code
- IaC scanner checks configurations
- Misconfigurations detected
- Deployment:
- Allowed
- Blocked based on policies